Grounded Exploration: Decoupled Planning and Terrain-Adaptive Execution from Onboard World Feedback

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Abstract

We study mapless real-world exploration from onboard world feedback. The target setting is a small rover in unknown and changing environments where external maps, privileged state, and simulator initialization are unavailable, and where the policy must be initialized and improved directly on hardware. Our current system decouples planning from execution: a policy selects a relative heading, a terrain-adaptive executor realizes motion, a Predictive Contextual Visual Memory (PCVM) backend supplies novelty, and the reward combines novelty with executed motion, coverage expansion, and penalties for zero-progress and revisits. Preliminary indoor runs already support multi-region exploration, while also exposing the present limitations: weak visual place discrimination, reward terms that still require calibration, and sensitivity to low-cost sensing. The working hypothesis for the final paper is that stronger visual representations and better reward alignment will shift the system from motion-dominated exploration toward genuinely visual novelty-seeking behavior.

1. Introduction & Background

Autonomous exploration is often framed as mapping: estimate free space, detect frontiers, and plan to informative boundaries (Yamauchi, 1997; Thrun et al., 2005). That framing weakens for small robots in damaged buildings, cluttered interiors, outdoor paths, or rubble, where sensing is weak, traversability changes over time, and the policy may need to start from random initialization on the real platform itself. We therefore study mapless exploration from world feedback: the robot improves online from the consequences reported by its own sensors and execution stack.

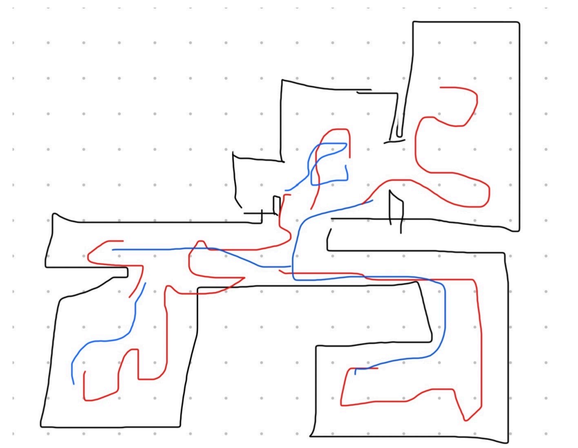

Figure 1. Placeholder first figure. Current best real run: corridor → room → similar second corridor → second room. Replace with a cleaned apartment map/trajectory figure.

The core hypothesis is that useful exploration does not require learning motor PWM. The policy should decide where to try next; a deterministic executor should decide how to realize that intent under uncertain terrain, actuator limits, and near-obstacle operation. This separation matters on hardware because raw motor RL entangles exploration credit assignment with contact handling, wheel slip, and low-level safety. Our current policy therefore outputs a scalar relative heading, and the executor drives forward until the front range sensor or safety layer terminates the action.

Intrinsic motivation is a natural reward source (Pathak et al., 2017; Burda et al., 2019; Savinov et al., 2019), but real robots expose failure modes that are easy to hide in simulation: blur can look novel, wall contact can create new pixels, and loops can accumulate reward without discovering meaningfully new places. We therefore gate novelty by executed motion, add coverage-expansion terms, and log every reward com-

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ponent so runs can be audited against video. Saved human and autonomous trajectories are used for analysis and reward debugging only, not for supervised policy pretraining. At present, the strongest part of the system is the grounded interface between heading selection, terrain-adaptive execution, and auditable reward; the longer-term target is to make stronger visual encoders and memory design materially influence action choice rather than leaving exploration dominated by proprioceptive and geometric proxies.

2. Methods

At time t, the rover receives $o_t = (I_t, d_t, m_t, a_{t-1}, e_{t-1})$: camera image, range readings, inertial/proprioceptive state, previous high-level action, and executor feedback. The policy selects $a_t \in [-1, 1]$, interpreted as a relative heading $\theta_t = 75°a_t$. The executor turns by θ_t and drives until the front threshold, max duration, or safety recovery stops the action. Dead-reckoned pose is logged only as a diagnostic, not treated as ground truth.

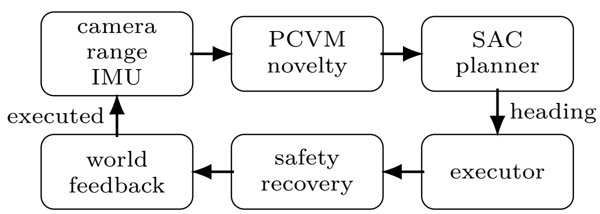

Figure 2. Current decoupled architecture. PCVM supplies novelty, SAC chooses heading, and executor/safety modules return grounded world feedback.

We call the current memory backend Predictive Contextual Visual Memory (PCVM). PCVM maps the frame and action context to an embedding z_t and stores representative embeddings. Novelty is computed from a weighted combination of memory distance and Random Network Distillation (RND; Burda et al., 2019), where RND compares a learned predictor against a fixed target network:

$$n_t = \alpha \,\mathrm{dist}(z_t, \mathcal{M}_{t-1}) + \beta \,\|f_{\mathrm{pred}}(I_t) - f_{\mathrm{target}}(I_t)\|_2^2. \quad (1)$$

The final encoder is not fixed; we have tested lightweight CNNs, recurrent path-conditioned memory, and frozen backbones.

Reward combines perceptual novelty with execution feedback:

$$r_t = r_t^+ - r_t^-, \quad (2)$$
$$r_t^+ = w_n \nVdash [M_t] n_t + w_c C_t + w_d \Delta s_t + w_s S_t, \quad (3)$$
$$r_t^- = \lambda_r R_t + \lambda_0 Z_t + \lambda_l L_t + \lambda_o O_t. \quad (4)$$

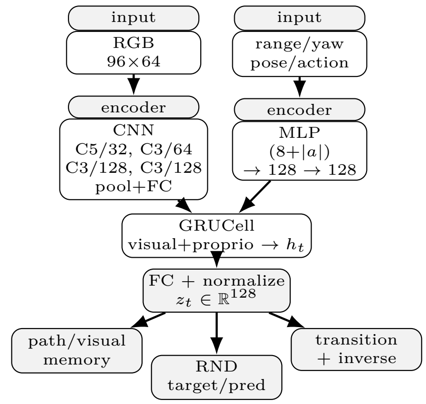

Figure 3. PCVM module: visual and proprioceptive features feed a recurrent latent with memory, RND, transition, and inverse-action heads.

Here M_t gates novelty by motion, C_t denotes cluster/coverage expansion, Δs_t executed distance, and R_t, Z_t, L_t, O_t recovery, zero-forward, revisit, and obstacle penalties. We currently use SAC (Haarnoja et al., 2018), with replay populated online from rover experience.

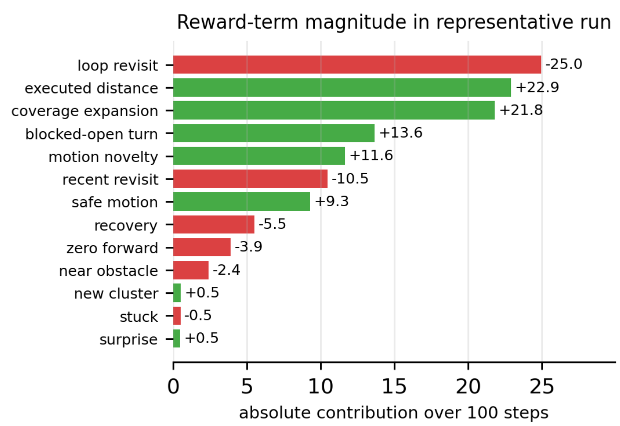

Figure 4. Absolute reward-term contributions in the representative 100-step run. This diagnostic shows which terms dominate the scalar objective and motivates reward-term ablations beyond hardware/executor ablations.

Temporary note, delete later. Layer two or three of these contribution charts from early/mid training and one from a later 5+ epoch continuation to see whether the dominant reward terms change order as the policy improves.

3. Preliminary Real-Rover Experiments

The motivating deployment is an unknown and changing structure, such as a small rescue rover entering a damaged building. The robot cannot rely on a precomputed map, but it can continuously evaluate the consequences of its actions through onboard feedback. The experiments reported here happen indoors because that is our current controllable testbed, not because the formulation is specific to apartment rooms. In the representative run shown in Figure 1, the current theta-front policy produced the sequence corridor → room → visually similar second corridor → second room.

The representative 100-step continuation run scored +31.5, traveled 20.3 m, reached 5.7 m net displacement, and expanded the dead-reckoned bounding box to 22.3 m^2. The same run also contained two contact/stall events and many recoveries. The result is therefore best read as a proof of usable multi-region traversal under weak sensing, not as a finished navigation system.

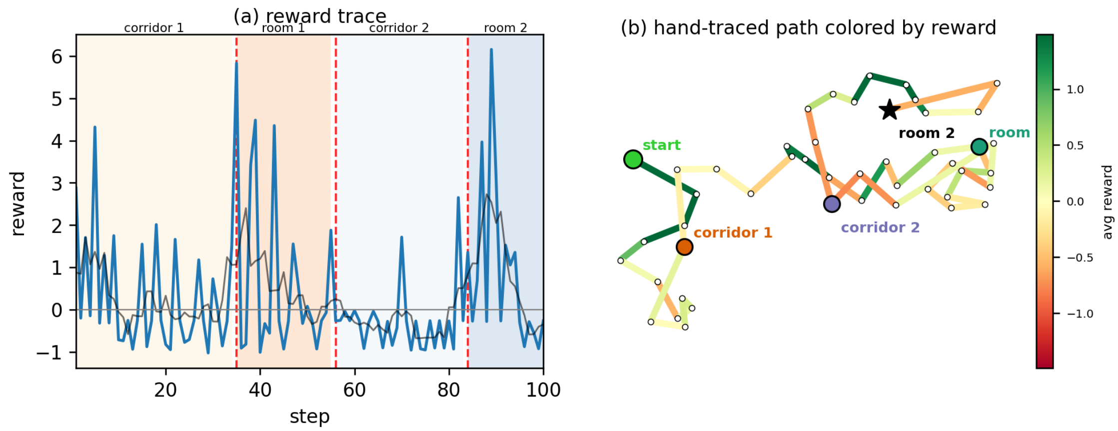

Figure 5. Placeholder reward trace and hand-traced qualitative path from the latest good real-rover run. The right panel is colored by average reward stretched over the drawn path segments and should be replaced with a cleaner map/trajectory from a stronger repeated run before final submission.

Memory / reward ablation	Return ↑	Travel ↑	BBox ↑	Visual novelty ↑	Zero-prog. ↓	Status
XYZ – Deterministic						baseline
Path memory only						planned
Visual memory only						planned
Path + visual memory						current
Path + visual + surprise						planned
+ coverage/open-turn reward						current
Full Grounded Exploration stack						current
Full stack + stronger vision encoder						planned
Perfect Human Reference						ceiling

Table 1. Planned ablation ladder and objective evaluation metrics. Empty cells mark repeated-run measurements to fill in later.

Temporary note, delete later. This table should test whether exploration comes from path/proprioceptive memory, visual memory, or their combination. The key question is whether visual novelty starts contributing once stronger encoders are used. Columns are objective run metrics: scalar return, estimated travel, dead-reckoned bounding-box area, visual novelty/cluster contribution, and zero-progress/already-blocked steps. Perfect Human Reference is the effortful manual ceiling, not a learned model.

4. Ablations

Current evidence points first to action interface: short local targets produced timid motion, while relative-heading plus world-bounded execution reached multiple rooms. This motivates a fixed-distance target vs. drive-until-front ablation. A second ablation is requested-action memory vs. executed-action memory, because safety clipping and recovery change what physically happened. A third is surprise on/off: transition error may capture useful unmodeled outcomes, but may also reward blur, noise, or collision artifacts.

The coverage reward was introduced after visual review showed that a two-room run scored lower than a corridor loop. Future ablations should rank frozen runs by human labels: corridor loop < corridor+one-room < corridor+two-room. Architecturally, the PCVM module still aliases similar rooms/corridors, so we plan to compare lightweight CNN features, pretrained encoders, and dual path/visual memory while keeping the same executor.

Table 2. Placeholder vision-backbone ablation.

Visual encoder	Signal to test	Status
Tiny PCVM CNN	current weak baseline	current
MobileNet	cheap pretrained features	planned
DINO-style encoder	semantic view similarity	planned
V-JEPA	predictive world features	offline

Temporary note, delete later. Decide which metric should define success for this table, then add at least one or two more vision/backbone alternatives before final submission.

5. Discussion

This draft supports a narrow claim: decoupled planning and terrain-adaptive execution can produce useful real-rover exploration from onboard world feedback. Future work is safety-aware reward, separate sensor/safety monitors for low-cost hardware, and stronger pretrained visual encoders for room-level novelty. The current results should not be over-interpreted until repeated runs and ablations replace the placeholder table.

References

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A. Appendix Placeholder

Extended run sheets, reward-audit details, and repeated-run ablations will go here.

B. Robot Platform

The robot is built around a Raspberry Pi 5, which serves as the primary computational unit responsible for sensor processing, decision-making, and execution of the machine learning algorithms. Power is supplied by an 18.5 V 4200 mAh 5S battery pack, enabling portable operation while maintaining sufficient energy density for both computation and motor control. The electrical architecture utilizes two dedicated step-down converters: the first provides a regulated 12 V rail capable of delivering up to 2 A for the dual DC drive motors through an L298N motor driver, while the second converter supplies stable 5 V Power Delivery to the Raspberry Pi system.

The robot integrates multiple sensing modalities to support autonomous navigation and environmental awareness. Positioning data is obtained through a GY-NEO6MV2 connected via UART communication, while orientation and motion tracking are handled by an MPU-9050 connected over the I^2C bus. Visual input is provided by a Pi Camera Module 3 attached through the MIPI interface, enabling real-time image acquisition for machine learning inference and navigation tasks. Additionally, three ultrasonic sensors positioned across the front and side sections of the robot provide short-range obstacle detection and spatial awareness. The chassis itself was fully custom-designed and manufactured using 3D printing techniques, allowing rapid iteration and mechanical optimization throughout the development process. The resulting system combines embedded computing, sensor fusion, and custom mechanical design into a compact robotic platform suitable for autonomous experimentation and adaptive learning applications.

Figure 6. Placeholder for final robot photograph.

Figure 7. Placeholder for final Raspberry Pi wiring diagram.